Learning-based Speculator (§3)

[inputs]

Ensembled Small Speculative Models (SSMs)

Fine-tuned SSM

Quantized LLM

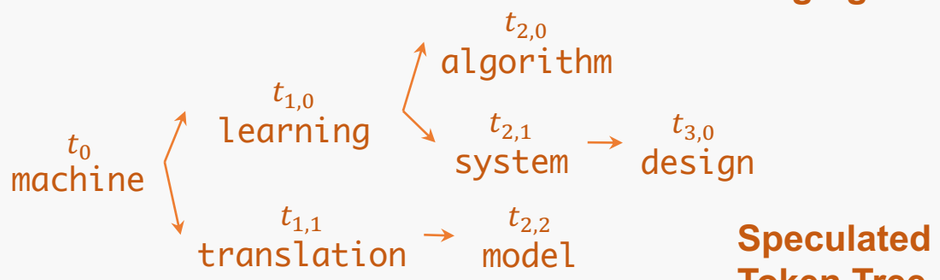
Distillated LLM

Knowledge Retriever

User-Defined Function

Output 0: [machine] learning algorithm
Output 1: [machine] learning system design
Output 2: [machine] translation model

Tree Merging



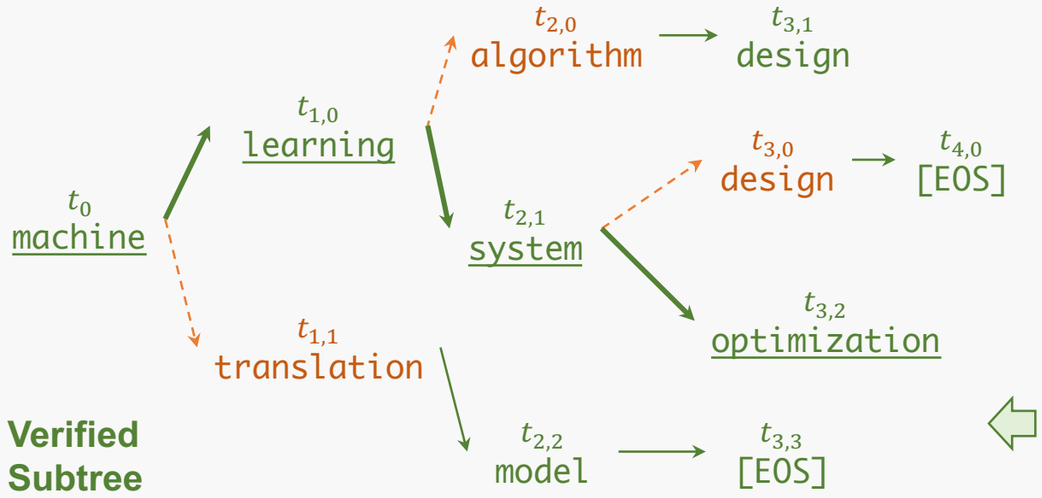
Speculated Token Tree

Verified output: machine learning system optimization

Synchronization

Linearization

Token Tree Verifier (§4)



Verified Subtree

[machine] → learning → algorithm system → design translation → model

Tree-based Parallel Decoding

LLM

learning system design optimization [EOS] model [EOS]